MILITARY SPECIFICATION

BOARD, COMPONENT: 10548362

1. SCOPE

1.1 This specification covers a component board used in Klystron Power Supply 10548336, a part of Radar Set AN/VPS-2. The component board provides circuitry for generating trigger pulses to form the modulator pulse. The component board also provides circuitry for overload sensing and relay closures for overload protection.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-C-46357	Computer, Gun Direction; M18 Manufacturing Processes and Inspection of

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F10548362	Board, Component

PACKAGING DATA SHEET

10548362	Packaging of Board, Component

(Copies of specifications, standards, drawings and packaging data sheets required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The component board shall be manufactured in accordance with Drawing F10548362 and drawings pertaining thereto.

3.2 General specifications. - Where applicable, the component board shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishings
d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

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3.2.2 MIL-C-46357.

a. Etched circuit assemblies
b. Wiring
c. Threaded fasteners
d. Marking
e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration. - The component board shall be mounted on the vibration table in normal service position. The component board shall meet the requirements of 3.4 and 3.5 after being subject to continuous vibration for 2 hours along each of three mutually perpendicular axes within the frequency range of 10 to 55 hertz (Hz) and a half amplitude excursion of 1/64 inch. The rate of change of frequency at each frequency range shall not exceed 1 Hz per second. Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock. - The component board shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G) and a pulse duration of 11 ±1 milliseconds (ms). The G value of each impulse shall be within ±10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5.5 ms.

3.3.3 Altitude.

3.3.3.1 Operating. - The component board shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Nonoperating. - The component board shall meet the requirements of 3.4 and 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. - The component board shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40° F and +155° F.

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3.3.4.2 Nonoperating (storage). - The component board shall not suffer permanent damage when subjected to not less than 4 hours at each temperature extreme of -65° F and +160° F, and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (nonoperating). - The component board shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity of 95 ±5 percent for a period of not less than 10 days.

3.4 Electrical. - Each conductor shall be electrically continuous as specified on the schematic diagram of the component board.

3.5 Performance.

3.5.1 Input and output characteristics. - The component board shall provide main and delayed gated trigger outputs synchronized with the radar system timing signals. The component board shall also provide HV power supply overload signals that will sense an overload condition, and shall latch up relays providing closures to lamp indicator circuits. When supplied with the inputs specified in Table I, the component board shall yeild the outputs specified in Table I.

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TABLE I. - INPUT AND OUTPUT CHARACTERISTICS

REQUIREMENT	CHARACTERISTICS	TERMINALS
Input voltage	+28 ±0.1 Vdc regulated, capacity 200 mA.	E19(+) E3(Ret)
	+6 Vdc ±0.1 Vdc regulated, capacity 150 mA min.	E13(+) E4(Ret)
	+6 Vdc ±0.1 Vdc regulated, capacity 150 mA min.	E9(+) E4(Ret)
Input reset voltage	+28 ±0.1 Vdc regulated, capacity 25 mA.	E15(+) E3(Ret)
Input signals, trigger	Amplitude: +25 ±1 V pulse. Pulse width: 0.6 ±0.1 μs. PRF: 20 ±1 kHz. Rise time: 0.03 μs max.	E5 E7 (shld.)
Output signals		
Main trigger output	Amplitude: 8 ±2 V pulse. Pulse width: 1.1 ±0.2 μs. PRF: 20 ±1 kHz. Rise time: Not less than 0.05 μs at +4.0 volt level.	E6 E3 (shld.)
Delayed trigger output	Amplitude: 9 ±1 V pulse. Pulse width: 1.4 ±0.2 μs. PRF: 20 ±1 kHz. Rise time: Not less than 0.05 μs at +6.5 volt level. Delay: 22 ±2 μs between leading edge of main trigger and leading edge of delay trigger.	E2 E4 (shld.)
Overload output signals		
HV overload	Relay closure, +28 ±1 Vdc.	E18
Mod. overload	Relay closure, +28 ± 1 Vdc	E11

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be listed in MIL-STD-109.

4.2 First article (initial production) submission. - The requirement for first article approval and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three component boards manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

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4.3 Inspection provisions.

4.3.1 Submission of product. - Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2. Examination and tests.

4.3.2.1 Components and subassemblies. - All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provision (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection. - Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in 4.3.2.3 and 4.4 herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in 4.3.2.3 and 4.4 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.3 Functional tests. - The requirements and tests in Table II shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature (60° F and 90° F).

TABLE II. - CLASSIFICATION OF DEFECTS
(100 percent inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
CRITICAL: NONE		
MAJOR:		
101. Electrical	3.4	4.6.2
102. Performance	3.5	4.6.3
MINOR: NONE		

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4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling. - One component board shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table III. Three component boards shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in Table IV.

TABLE III. - SPECIAL SAMPLING

	CONTROL TEST	REQUIREMENT	TEST PROCEDURES
301.	Fabrication	3.1	Applicable drawings - Visual
302.	General specifications	3.2	MIL-F-13926 & MIL-C-46357 - Visual

TABLE IV. - SPECIAL SAMPLING ENVIRONMENTAL

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Vibration	3.3.1	4.6.1.1
304.	Shock	3.3.2	4.6.1.2
305.	Altitude	3.3.3	4.6.1.3
306.	Temperature	3.3.4	4.6.1.4
307.	Humidity (nonoperating)	3.3.5	4.6.1.5

4.4.1 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

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4.5.1 Government-furnished inspection equipment. - Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

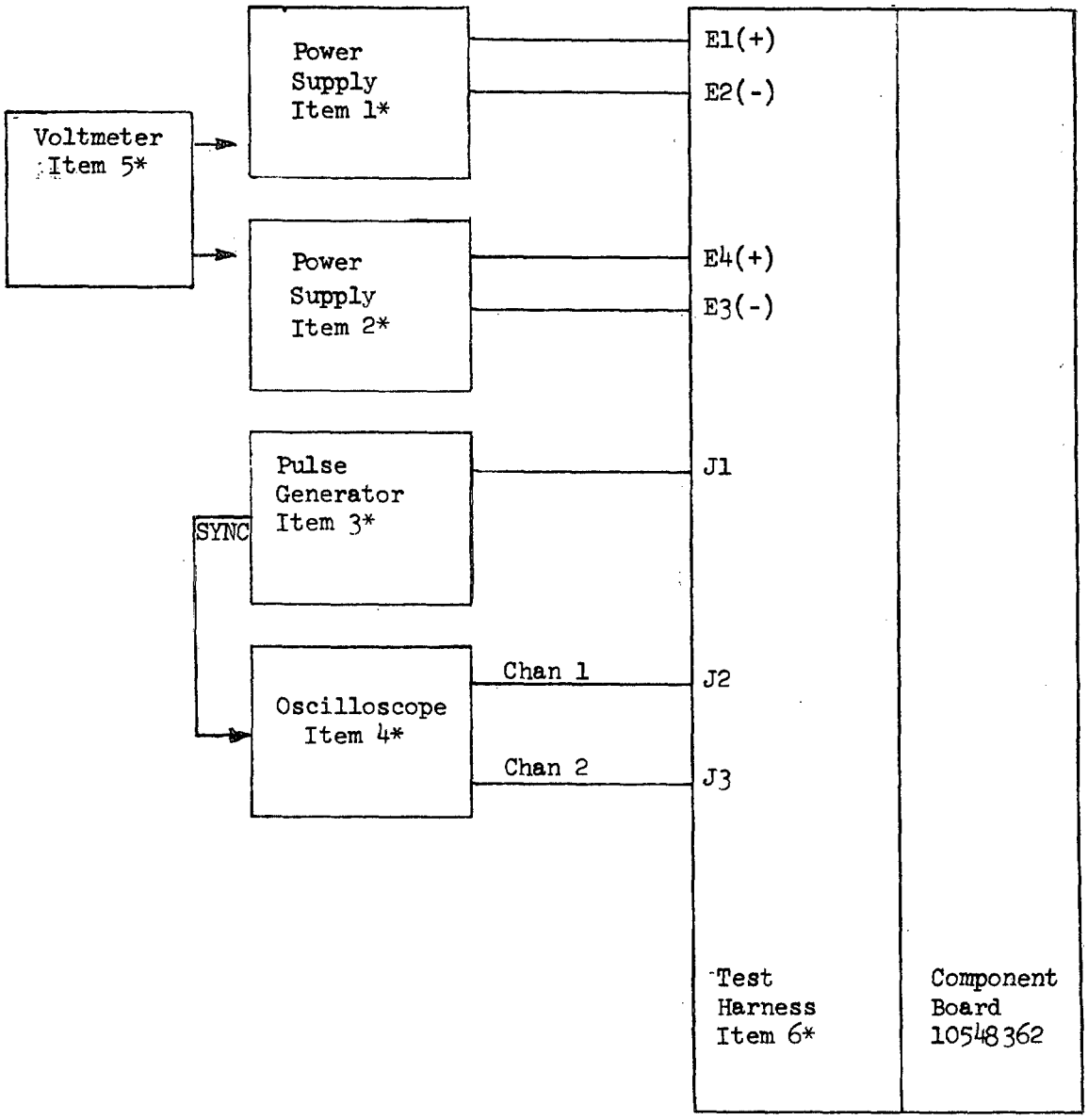
4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4.5.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Test equipment - In conjunction with 4.5.2.2, the standard test equipment listed in Table V shall be utilized in performing the applicable tests of 4.6. (See Figure 1).

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*Refer to 4.5.2.3

FIGURE 1. COMPONENT BOARD TEST SETUP.

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TABLE V. - TEST EQUIPMENT LIST

ITEM NUMBER		DESCRIPTION
1.	Power supply	0 to 30 Vdc, regulated; ac ripple less than 50 mV peak-to-peak; 500 mA current capacity.
2.	Power supply	0 to 10 Vdc, regulated; ac ripple less than 50 mV peak-to-peak; 500 mA current capacity.
3.	Pulse generator	Minimum positive pulse amplitude: +30 V. Pulse width: adjustable from 0.2 to 20 μs. Rise time: 20 μs max. PRF: 20 kHz.
4.	Oscilloscope	Dual channel. Vertical rise time: 50 μs max. Sweep lengths: 1.0 μs, 10 μs and 50 μs. Scope probe: 10 megohm input resistance.
5.	Voltmeter	0 to 50 Vdc, accuracy 0.1% of reading.
6.	Test harness	See Figure 2. See Table VI for test harness parts list.

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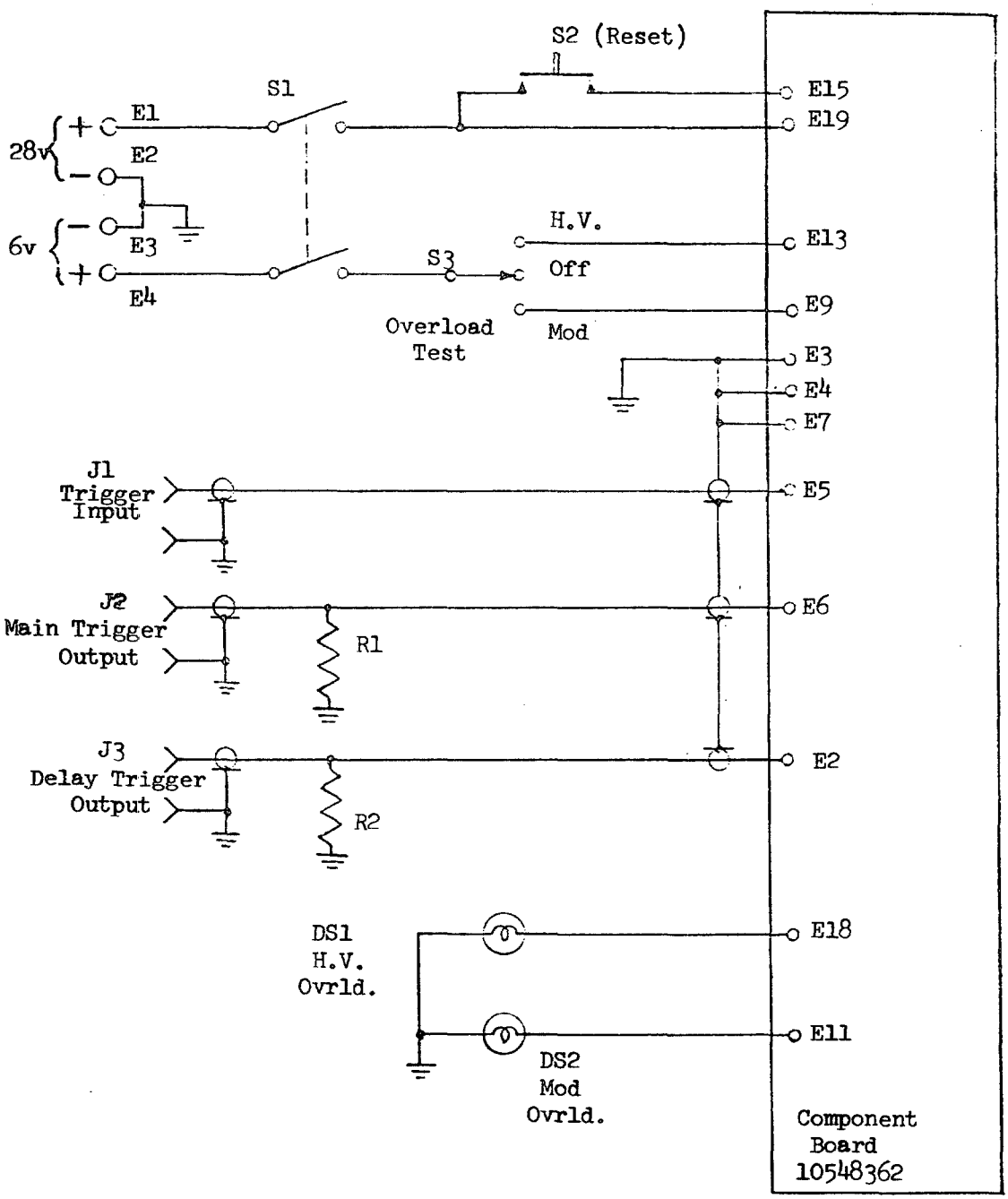


FIGURE 2. TEST HARNESS

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TABLE VI. - TEST HARNESS PARTS LIST

DESIGNATION	PART	DESCRIPTION
DS1, DS2	Lamp and cartridge	28 V at 40 mA.
E1, E4	Binding post	E.F. Johnson, 111-102 (red)
E2, E3	Binding post	E.F. Johnson, 111-103 (black)
J1, J2, J3	Connector	UG290A, BNC.
R1	Resistor	RC32GF100J
R2	Resistor	RC32GF300J
S1	Switch	DPST
S2	Switch	Pushbutton, momentary SPST NO.
S3	Switch	SPDT center off

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4.6 Test methods and procedures.

4.6.1 Environmental. - Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

4.6.1.1 Vibration. - The test shall be performed as specified in Method 514, Procedure IX, Part I, of MIL-STD-810 except that failure criteria do not apply. After vibration, the component board shall be tested as specified in 4.6.2 and 4.6.3 to determine compliance with 3.3.1.

4.6.1.2 Shock. - This test shall be performed as specified in Procedure I of Method 516 of MIL-STD-810 except that shock pulses shall be 11.0 ms and 15 G. After the shock impulses, the component board shall be tested as specified in 4.6.2 and 4.6.3 to determine compliance with 3.3.2.

4.6.1.3 Altitude (nonoperating and operating). - This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that in Step 2 the altitude shall be limited to 40,000 feet (5.5 inches Hg). Operation of Step 3 shall be the performance tests specified in 4.6.3.

4.6.1.4 Temperature.

4.6.1.4.1 High (operating and nonoperating). - This test shall be performed as specified in Procedure I, Method 501, of MIL-STD-810. The temperature of Step 2 shall be +160° F. The temperature of Step 3 shall be maintained for not less than 4 hours. The temperature of Step 4 shall be +155° F. In Step 5 the temperature shall be maintained for 4 hours and the component board shall be tested in accordance with 4.6.2 and 4.6.3 during this time.

4.6.1.4.2 Low (operating and nonoperating). - This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-810. The temperature of Step 2 shall be -65° F maintained for not less than 4 hours. The temperature of Step 4 shall be -40° F. In Steps 5 and 7, the component board shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.5 Humidity (nonoperating). - This test shall be performed in accordance with Procedure II, Method 507 of MIL-STD-810. The component board shall be tested in accordance with 4.6.3.

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4.6.2 Electrical. - Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.6.3 Performance. - The selected board assemblies shall be tested as follows:

4.6.3.1 Input and output characteristics. - Connect test equipment items 1 through 6 as shown in Figure 1. Turn switches S1 and S3 of test harness, item 6, to the OFF position. Connect the pulse generator, item 3, to the oscilloscope, item 4, and set up the pulse generator output as specified in Table I. Adjust the power supply voltages as specified in Table I.

4.6.3.2 Main trigger output. - Switch the oscilloscope, item 4, to channel 1 and verify that the output pulse has the following characteristics:

PRF: 20 kHz
Amplitude: +8 ±2 V
*Width: 1.1 ±0.2 μs
Rise time: 0.05 μs max.

4.6.3.3 Delay trigger output. - Switch the oscilloscope, item 4, to channel 2 and verify that the delay output pulse has the following characteristics:

PRF: 20 kHz
Amplitude: +9 ±1 V
*Width: 1.4 ±0.2 μs
Rise time: 0.05 μs

*If either width of 4.6.3.2 or 4.6.3.3 is out of required specification it will be necessary to select resistors for R22 and/or R33. For resistor values refer to parts list, P/L 10548362.

4.6.3.4 Delay performance test. - Connect the oscilloscope, item 4, to external sync between main trigger output and switch to alternate mode sweep. Verify that the delay between main trigger rise time and the delay trigger rise time is 24 ±2 μs.

4.6.3.5 High voltage overload. - Recheck the +28 V and the +6 V inputs. Switch S3 on test harness, item 6, to HV position and verify that the HV indicator lamp, DS-1, is lit and remains lit. Turn S3 on test harness, item 6, to the OFF position. Push S2 on item 6, reset button and verify that the HV overload indicator DS-1 goes out and remains out.

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4.6.3.6 Modulator overload.

4.6.3.6.1 Switch S3 on test harness, item 6, to the MOD position, and verify that the modulator overload indicator lamp DS-2 is lit and remains lit. Turn switch S3 on test harness, item 6, to the OFF position.

4.6.3.6.2 Push switch S2 on item 6, reset button and verify that the modulator overload indicator DS-2 goes out and remains out.

4.6.3.6.3 Switch S3 on test harness, item 6, to MOD, then to HV and then to OFF and verify that both DS-1 and DS-2 are lit and remain lit.

4.6.3.6.4 Reset switch S2 on test harness, item 6, and verify that both indicators go out and remain out. Repeat tests several times.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10548362. The level of protection shall be as specified in the procurement document.

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6. NOTES

6.1 Intended use. - The component board is an intergal board assembly within the klystron power supply, a subassembly of the Radar Set AN/VPS-2. The component board produces the main and delayed trigger outputs with respect to the timing synchronizing signals to produce the modulator output pulse. The component board also provides the necessary circuitry and relay closures for the protection of the high voltage and modulator when an overload is sensed.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1285-A256

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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